CBE 30355 Transport I Problem Set 8

Due via Gradescope on Thursday 11/5/26

For **each** of the problems you must attempt the problem yourself and discuss your approach with a classmate before using an AI assist.

For **each** problem solution indicate either that you didn't use AI or provide the name of a study buddy you discussed your solution approach with first!

What you should learn from problem set 8

- **Problem 1: Lubrication: defined flow rate.**
  - This is a very nice example of a defined flow rate lubrication problem, although you will need to be careful in applying your boundary conditions!
  - Reinforces that, as in all lubrication problems, the velocity profile across the thin direction is just a parabola... Thus, if you get the three constants by applying boundary conditions, symmetry, etc., the problem becomes much simpler!
  - More practice rendering equations dimensionless!
- **Problem 2: Lubrication flow between two disks**
  - This makes you work through the notes on defined flow rate lubrication problems.
  - Demonstrates how you can use the results of one problem to provide information on a rather different one (e.g., tying the results together).
  - Recognizing that sometimes "push button instruments" can yield systematic errors due to unanticipated effects, and thinking through how you could fix the results by proper recognition of why the results were in error...
- **Problem 3: Scaling & Thermodynamics**
  - This problem is a useful example of applying your intuition to get insight into how a real problem (e.g., a chimney) depends on various parameters.
  - Helps to see how thermo and transport are interrelated.
- **Problem 4: Dimensional Analysis**
  - Another example of scaling, this time using strict dynamic similarity.
  - Application of scaling to a real-world modeling problem.

Problem 1 (10 points)

Lubrication Layer Between Rollers

Consider the geometry below. Two sheets are drawn with a velocity U between two rollers of radius R whose surfaces are separated by a distance $2b_0$. The idea is to fill the space between the sheets with a viscous fluid of viscosity μ, and all the action takes place in a lubrication layer between the rollers, at least in the limit $b_0/R \ll 1$. Here we analyze this problem.

a. By scaling the flow equations in the gap (use Cartesian coordinates!), show how the force per width on the rollers F/L scales with the separation distance between the rollers and the other parameters of the problem.

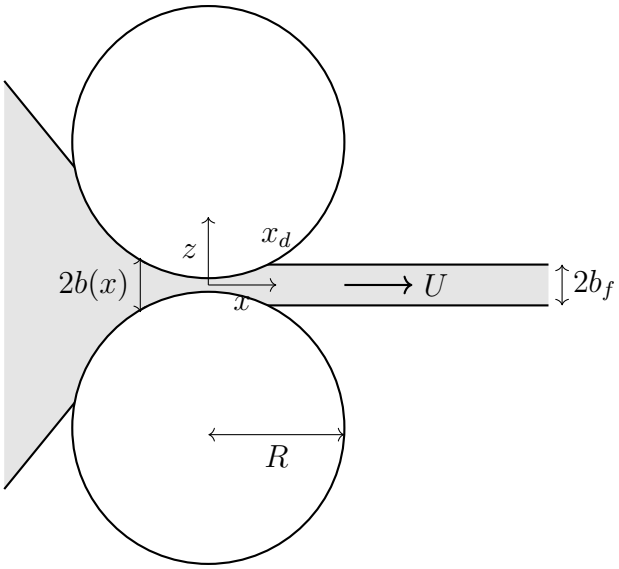
b. The sheet sandwich detaches from the rollers downstream at a point x_d and final gap width $2b_f > 2b_0$ when the pressure again returns to zero. Develop an implicit integral relation for the ratio b_f/b_0, and show that it doesn't depend on any of the other parameters of the problem.

c. Develop an integral relationship for the dimensionless pressure and force and evaluate it numerically.

Hint: The gap geometry in the lubrication limit is given by

$$b(x) = b_0 + \frac{x^2}{2R}$$

where x is the distance along the gap from the point of minimum separation. By continuity, the flow rate per extension into the paper (Q/W) through any vertical plane must be the same, and must equal the exit flow rate $U \cdot 2b_f$. Thus, this is a defined flow rate problem which makes things *much* easier!



Problem 2 (10 points)

Zero Error in Parallel-Plate Viscometers

In an interesting paper (Davies & Stokes, *J. Rheol.* 2005) a fundamental problem with current parallel-plate viscometers was identified. As you know, the viscosity of a fluid is measured in a parallel-plate viscometer by rotating one plate at a known angular velocity and measuring the resultant torque on the other plate. The viscosity calculation, however, requires accurate knowledge of the gap width between the plates.

In a modern instrument it is possible to set the gap accurately automatically *provided* you have the reference point of where the two plates are in contact: e.g., where the gap is zero. For an instrument such as this, you just push a button, the upper plate descends at a constant velocity of $100\,\mu$m/s, and the gap is considered to be zero when the measured upward thrust on the upper plate exceeds 10^4 dynes, providing the reference point for all subsequent gaps—very convenient! This would be fine if there were no fluid in the gap (e.g., a vacuum), but as you know, the viscosity of air is non-zero (1.8×10^{-4} poise). In this problem we analyze what the effect of air is on the gap zeroing of 3 cm radius parallel-plates.

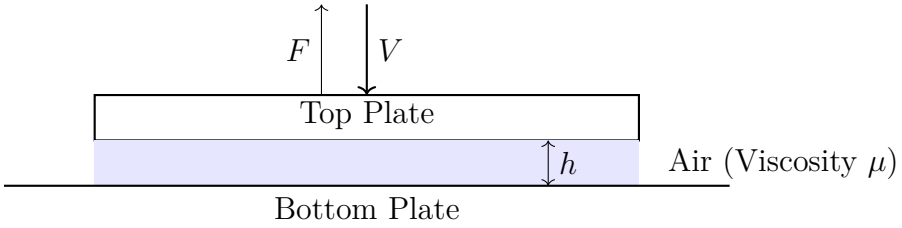
a. For a normal force detection threshold F, approach velocity V, air viscosity μ, and plate radius R, calculate the error in the gap set zero.

b. For the numbers given above, calculate the numerical value of the error in microns. Is it practical to reduce this error by an order of magnitude by decreasing V? Be specific!

c. If we set the gap to $250\,\mu$m using this incorrect zero (e.g., actual gap differs from $250\,\mu$m due to the zero error), what is the corresponding error in the viscosity measurement

μ_{exp}/μ of a fluid?

d. How can you use the output of the instrument to get the correct viscosity? Be specific! *(Hint: It will require analysis of a couple of measurements.)*



Problem 3 (10 points)

Thermodynamics and Scaling Analysis

Consider the temperature and fluid velocity in a chimney as a function of various physical parameters. The chimney has cross-sectional area A and height H, so the total volume of air in the chimney is AH. A constant energy source at the bottom supplies energy at a rate Q, uniformly over the base. The system is idealized: we neglect friction and heat loss and assume the energy input is independent of flow rate. This energy input heats the air, which expands and rises. Using conservation of energy and momentum, estimate the mass flow rate and the resulting temperature in the chimney.

a. For small values of Q, the temperature and density changes are small. What are the mass flow rate and chimney temperature in this limit?

b. For large values of Q, the density of the gas in the chimney is much less than that of the surrounding air. What are the mass flow rate and temperature in this limit?

Hint: You will need to recall some concepts from Thermodynamics.

Problem 4 (10 points)

Modeling Air Entrainment in a Draining Tank

The use of rail cars to ship heavy crude down from Canada or from the fields of North Dakota has exploded (sometimes literally) in recent years. As a consequence of safety requirements, tank cars are being redesigned. You are assigned the task of performing a scale model test to determine the maximum drainage rate.

Consider the liquid tanker rail car depicted below. The 53 foot long tanker contains heavy

crude with viscosity $\mu = 10$ poise and density $\rho = 0.922\,\mathrm{g/cm^3}$. As the tank drains, a vortex will form and eventually air will be entrained into the drainpipe—something we want to avoid. We want to determine the allowable operating conditions for our large tank by studying the behavior of a geometrically similar model system, but using water as the working fluid.

a. For strict dynamic similarity, what should be the scale-down ratio of the scale model?

b. How does the draw-off rate (volumetric flow rate) scale between the model and full-size tanker?

